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(71) Applicant: TEVA PHARMACEUTICALS INTERNATIONAL GMBH [CH/CH]; SCHLUSSELSTRASSE 12, 8645 JONA (CH).

(71) Applicant (for BB only): TEVA PHARMACEUTICALS USA, INC. [US/US]; 1090 HORSHAM ROAD, P.O. BOX 1090, NORTH WALES, Pennsylvania 19454-1090 (US).

(72) Inventors: KATSAV, Luna Ben-Sahel; Pinchas Meiri 4, Petah Tikva (IL). GOLDSHTEIN, Jenny; Ha-Gilad 1, Netanya (IL). ADANI, Limor; Rivka Guver 3, Petah Tikva (IL). MASARWA, Abed; Kareem El- Jebaly, Tay-ibe (IL). MUTHUSAMY, Anantha Rajmohan; 104 Amman Kovilpatti South Street, 626123 Sivakasi, Tamil Nadu (IN). SOMASUNDARAM, Meenakshi Sundaram; 63 Kamatchiyar Street, Sattur, 626203 Virudhu Nagar DT, Tamilnadu (IN). MUPPALLA, Siva Rama Krishna; C102 Paramount Golf Foreste, Opposite Zeta sector-1, 201311 Greater Noida, Uttar Pradesh (IN). PANDEY, Anand Kumar; Vill: Chhattupatti, PO-Dhingwas, 230128 Pratapgarh, Uttar Pradesh (IN). SONAR, Jitendra Kamalakar; 9-A Hridayam, Khanderao Nagar, Lonkheda, Tal: Shahada, 425409 Dist:- Nandurbar, Maharashtra (IN). KUMAR, Sumit; Vill:Muzafat Khurd, PO:Muzafat, Kalan, 135021 Dist:Yamunanagar, Haryana (IN).

(74) Agent: BREW, Michael R. et al.; Carter, DeLuca & Farrell LLP, 576 Broad Hollow Road, Melville, New York 11747 (US).

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— as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))

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(54) Title: SOLID STATE FORMS OF VERICIGUAT AND PROCESS FOR PREPARATION THEREOF

(57) Abstract: The present disclosure encompasses solid state forms of Vericiguat, in embodiments crystalline polymorphs of Vericiguat, processes for preparation thereof, and pharmaceutical compositions thereof.

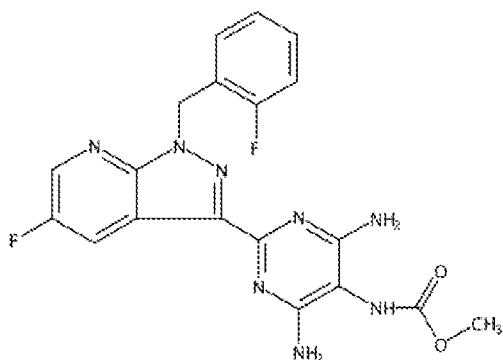
SOLID STATE FORMS OF VERICIGUAT AND PROCESS FOR PREPARATION THEREOF

FIELD OF THE INVENTION

[0001] The present disclosure encompasses solid state forms of Vericiguat, in embodiments crystalline polymorphs of Vericiguat, processes for preparation thereof, and pharmaceutical compositions thereof.

BACKGROUND OF THE INVENTION

[0002] Vericiguat has the chemical name Methyl {4,6-diamino-2-[5-fluoro-1-(2-fluorobenzyl)-1H-pyrazolo[3,4-b]pyridin-3-yl]pyrimidin-5-yl} carbamate. Vericiguat has the following chemical structure:



[0003] Vericiguat is a soluble guanylate cyclase (sGC) stimulator and is approved for reducing the risk of cardiovascular death and heart failure (HF) hospitalization following a hospitalization for heart failure or need for outpatient IV diuretics, in adults with symptomatic chronic HF and ejection fraction less than 45%.

[0004] Vericiguat is disclosed in WO 2011/147809 (referred to herein as WO '809). WO'809 also discloses hydrochloride, sulphate, phosphate, mesylate, ethane-1,2-disulphonate, maleate and nitrate salts of Vericiguat. WO2013/076168 discloses solid state forms, a di-dimethyl sulphoxide solvate, a dimethylformamide/water solvate, a triacetic acid solvate, a monohydrate and a dihydrate of Vericiguat.

[0005] Polymorphism, the occurrence of different crystal forms, is a property of some molecules and molecular complexes. A single compound, like Vericiguat, may give rise to a

An X-ray Powder Diffractogram (XRPD) of Vericiguat as per Example-1, Step-1.

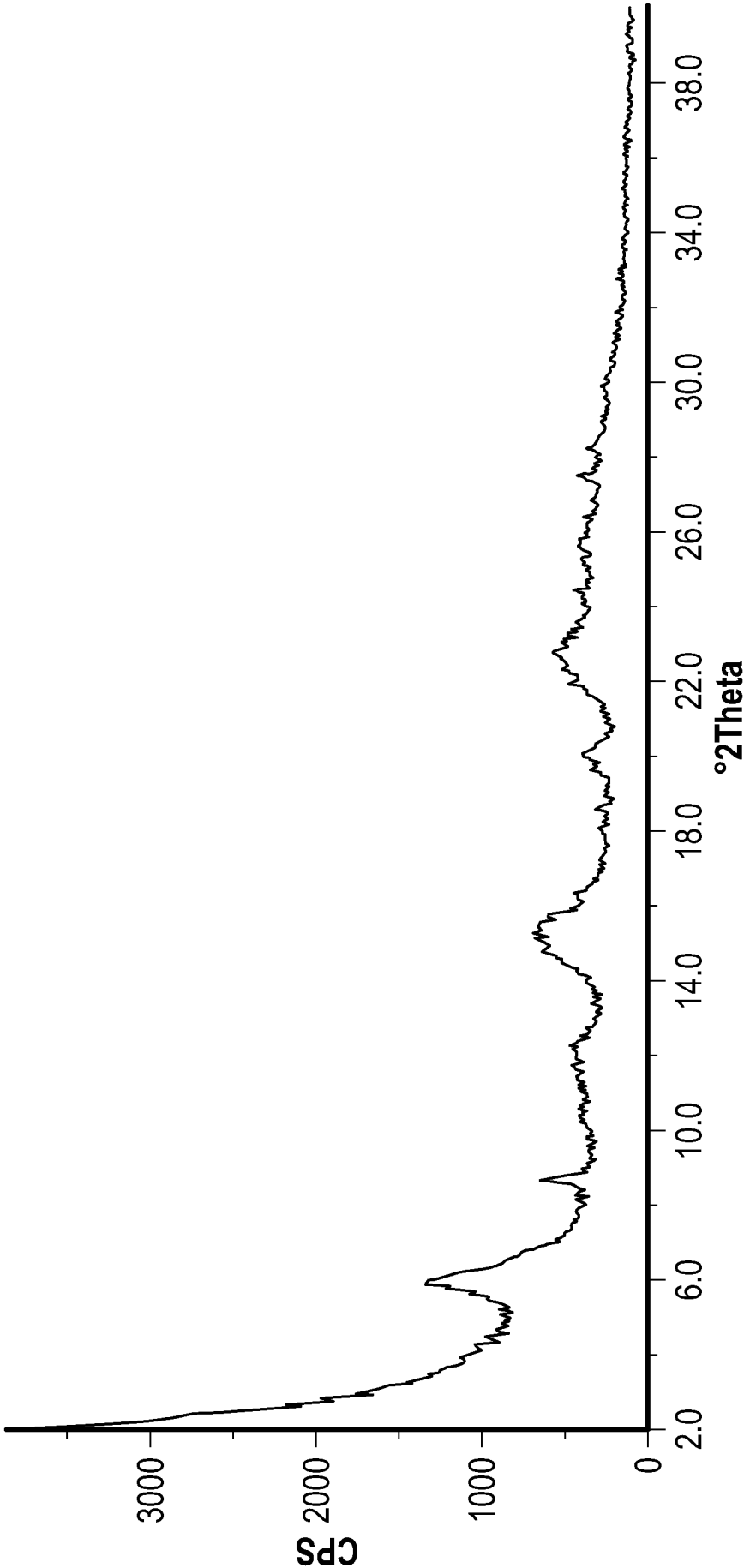


FIG. 7